DIP Practical — 9 Short MATLAB Programs

1. Image Enhancement Techniques

```
I=imread('logo.png');
if size(I,3)==3, I=rgb2gray(I); end
subplot(2,2,1); imshow(I); title('Original');
subplot(2,2,2); imshow(imadjust(I));
title('Adjusted');
subplot(2,2,3); imshow(histeq(I)); title('Histogram
Equalization');
subplot(2,2,4); imshow(adapthisteq(I));
title('Adaptive Equalization');
```

2. Histogram Equalization

```
I=imread('diphist1.jpg');
if size(I,3)==3, I=rgb2gray(I); end
H=histeq(I);
subplot(2,2,1); imshow(I); title('Original');
subplot(2,2,2); imshow(H); title('Equalized');
subplot(2,2,3); imhist(I); title('Original
Histogram');
subplot(2,2,4); imhist(H); title('Equalized
Histogram');
```

3. Image Restoration

```
I=im2double(imread('building.jpg'));
P=fspecial('motion',21,11);
B=imfilter(I,P,'conv','circular');
nsr=0.0001/var(I(:));
R=deconvwnr(B,P,nsr);
subplot(1,3,1); imshow(I); title('Original');
subplot(1,3,2); imshow(B); title('Blurred');
subplot(1,3,3); imshow(R); title('Restored');
```

4. Image Filtering

```
I=imread('crow.jpg');
if size(I,3)==3, I=rgb2gray(I); end
N=imnoise(I,'salt & pepper',0.05);
F=medfilt2(N,[3 3]);
subplot(1,3,1); imshow(I); title('Original');
subplot(1,3,2); imshow(N); title('Noisy');
subplot(1,3,3); imshow(F); title('Median Filtered');
```

5. Edge Detection

```
I=imread('dipedge.jpg');
if size(I,3)==3, I=rgb2gray(I); end
subplot(2,3,1); imshow(I); title('Original');
subplot(2,3,2); imshow(edge(I,'sobel'));
title('Sobel');
subplot(2,3,3); imshow(edge(I,'canny'));
title('Canny');
subplot(2,3,4); imshow(edge(I,'prewitt'));
title('Prewitt');
subplot(2,3,5); imshow(edge(I,'roberts'));
title('Roberts');
```

6. Image Compression

```
I=imread('mage3.jpg');
if size(I,3)==3, I=rgb2gray(I); end
n=5; w='sym8';
[C,L]=wavedec2(I,n,w);
X=wdencmp('gbl',C,L,w,n,30,'h',1);
subplot(1,2,1); imshow(I); title('Original');
subplot(1,2,2); imshow(X,[]); title('Compressed');
```

7. Image Subtraction

```
I=imread('nature.jpg');
if size(I,3)==3, I=rgb2gray(I); end
B=imopen(I,strel('disk',70));
R=imsubtract(I,B);
subplot(1,2,1);imshow(I);title('Original');
subplot(1,2,2);imshow(R);title('Subtracted');
```

8. Boundary Extraction

```
I=imread('flower.jpg');
if size(I,3)==3, I=rgb2gray(I); end
E=imerode(I,strel('disk',4,0));
B=imsubtract(I,E);
subplot(1,2,1);imshow(I);title('Original');
subplot(1,2,2);imshow(B,[]);title('Boundary');
```

9. Image Segmentation

```
I=imread('dipbrain.jpg');
if size(I,3)==3, I=rgb2gray(I); end
BW=imbinarize(I);
D=bwdist(~BW);
W=watershed(-D);
R=BW & (W~=0);
subplot(2,2,1);imshow(I);title('Original');
subplot(2,2,2);imshow(BW);title('Binary');
subplot(2,2,3);imshow(label2rgb(W));title('Watershed')
;
subplot(2,2,4);imshow(R);title('Segmented');
```

Last-minute memory map

- Enhancement — imadjust, histeq, adapthisteq
- Histogram — histeq, imhist
- Restoration — fspecial, imfilter, deconvwnr
- Filtering — imnoise, medfilt2
- Edges — edge
- Compression — wavedec2, wdencmp
- Subtraction — imopen, imsubtract
- Boundary — strel, imerode
- Segmentation — imbinarize, bwdist, watershed

Short Abstracts – 9 DIP MATLAB Programs

1. Image Enhancement: To improve image quality using intensity adjustment, histogram equalization, and adaptive histogram equalization.

2. Histogram Equalization: To improve image contrast by redistributing pixel intensity values using `histeq` and display the histogram using `imhist`.

3. Image Restoration: To restore a degraded image by applying motion blur and Wiener filtering.

4. Image Filtering: To remove salt-and-pepper noise from an image using median filtering.

5. Edge Detection: To detect image edges using Sobel, Canny, Prewitt, and Roberts operators.

6. Image Compression: To compress an image using wavelet decomposition and thresholding techniques.

7. Image Subtraction: To perform morphological opening and subtract the processed image from the original image.

8. Boundary Extraction: To extract the boundary of an object using morphological erosion and image subtraction.

9. Image Segmentation: To divide an image into meaningful regions using distance transformation and the watershed algorithm.